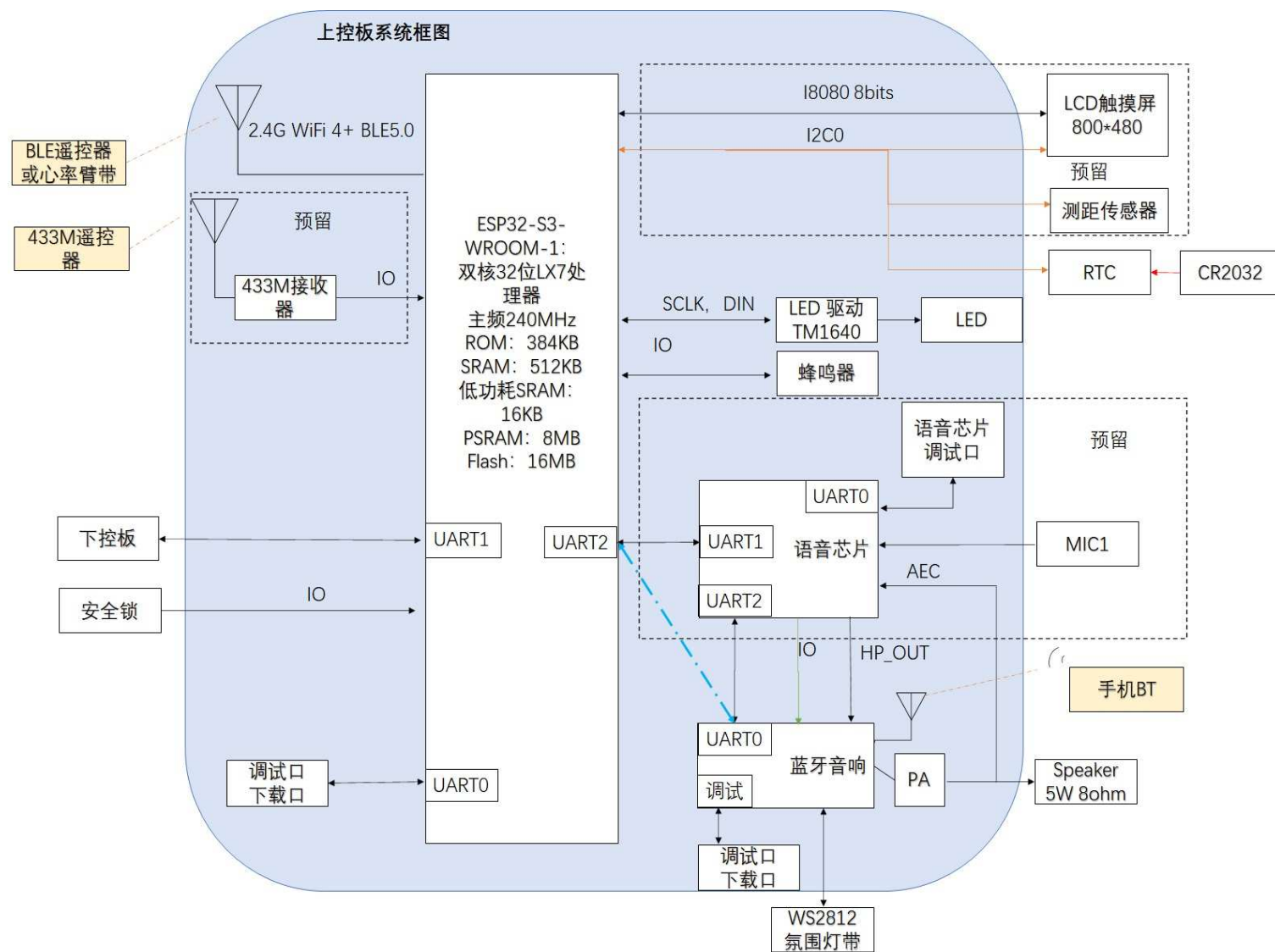


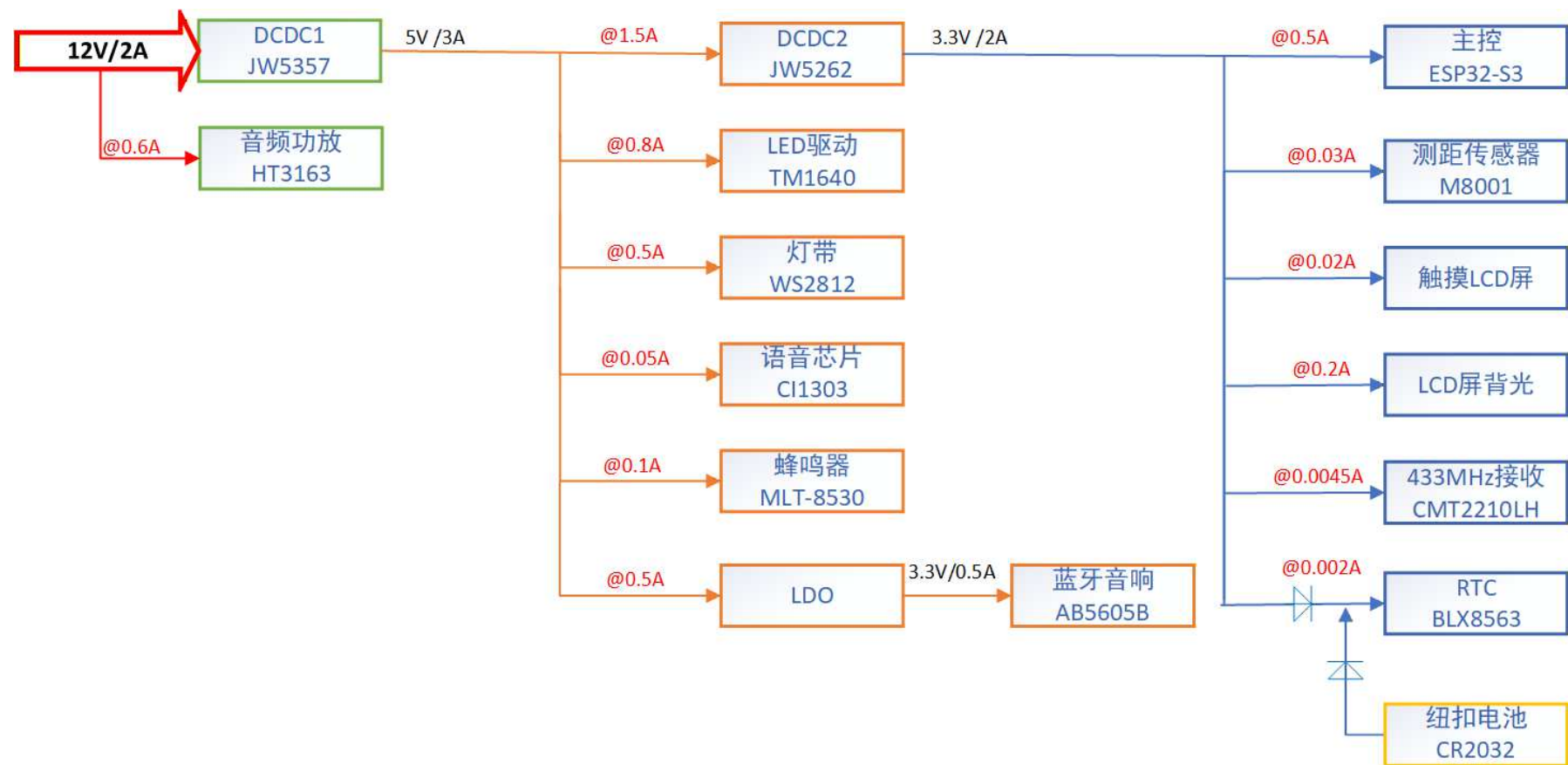
BLOCK



V0.1

Zhejiang Mairuike Technology Co., Ltd.			
Design Name			
T28_Upper Control Board			
Size	Page Name	Rev	
A3	BLOCK	<V01>	
Date	Thursday, June 18, 2026	Sheet	1 of 10

POWER TREE



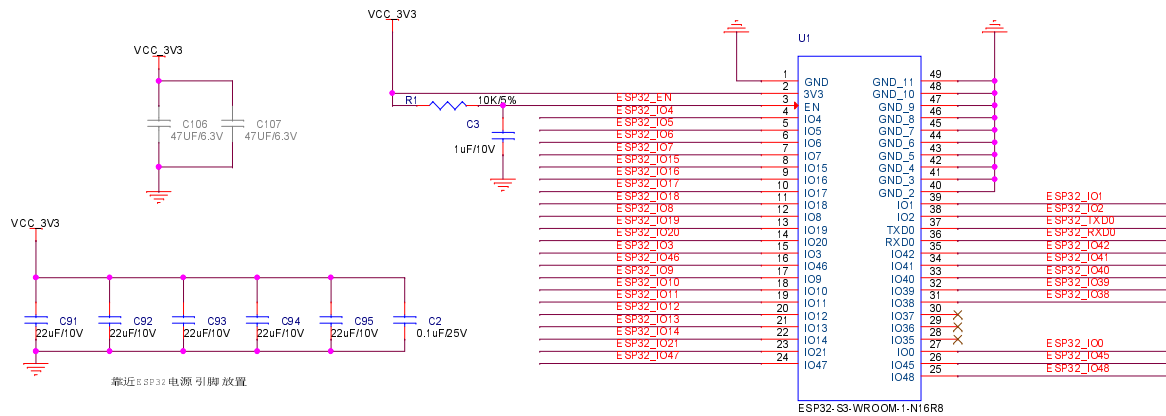
V0.1

 Zhejiang Mairuike Technology Co., Ltd.			
Design Name: T28_Upper Control Board			
Size: A3	Page Name: POWER TREE		Rev: <V01>
Date: Thursday, June 18, 2026	Sheet: 2	of 10	

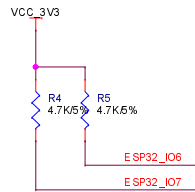
分页标识

LCD_CS_n	ESP32_IO48	TP19
LCD_RS	ESP32_IO45	TP20
LCD_WR_n	ESP32_IO0	TP21
LCD_RD_n	ESP32_IO38	TP22
LCD_RST_n	ESP32_IO39	TP23
LCD_D0	ESP32_IO10	TP24
LCD_D1	ESP32_IO9	TP25
LCD_D2	ESP32_IO11	TP26
LCD_D3	ESP32_IO12	TP27
LCD_D4	ESP32_IO13	TP28
LCD_D5	ESP32_IO14	TP29
LCD_D6	ESP32_IO21	TP30
LCD_D7	ESP32_IO47	TP31
M8001_INT	M8001_INT	TP32
TP_INT	TP_INT	TP33
M8001_RST	M8001_RST	TP34
TP_RST	TP_RST	TP35
WS2812_DIN	WS2812_DIN	TP36
PA_CTRL	ESP32_IO6	TP37
I2C_SDA	ESP32_IO6	TP38
I2C_CLK	ESP32_IO7	TP39
ESP_TX1	ESP32_IO17	TP40
ESP_RX1	ESP32_IO18	TP41
TM1640_LED_DIN	TM1640_LED_DIN	TP42
ESP_TX2	ESP32_IO19	TP43
ESP_RX2	ESP32_IO20	TP44
TM1640_LED_SCK	TM1640_LED_SCK	TP45
433M_DIN	ESP32_IO42	TP46
SAFETY_IN	ESP32_IO41	TP47
BL_PWM	ESP32_IO40	TP48
PWR_KEY	ESP32_IO1	TP49
BZ_OUT	ESP32_IO2	TP50
BT_WS2812_DIN	BT_WS2812_DIN	TP161

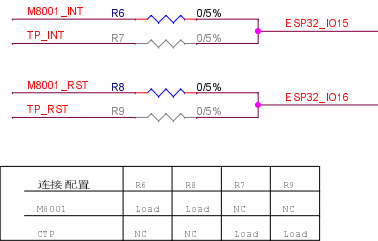
ESP32-S3



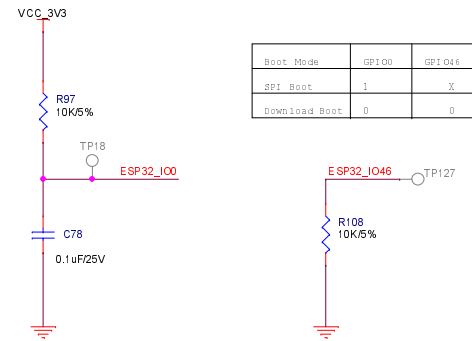
I2C
上拉电阻



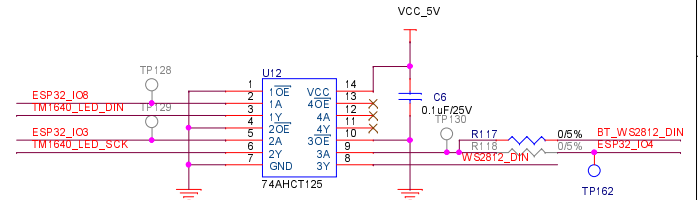
引脚复用



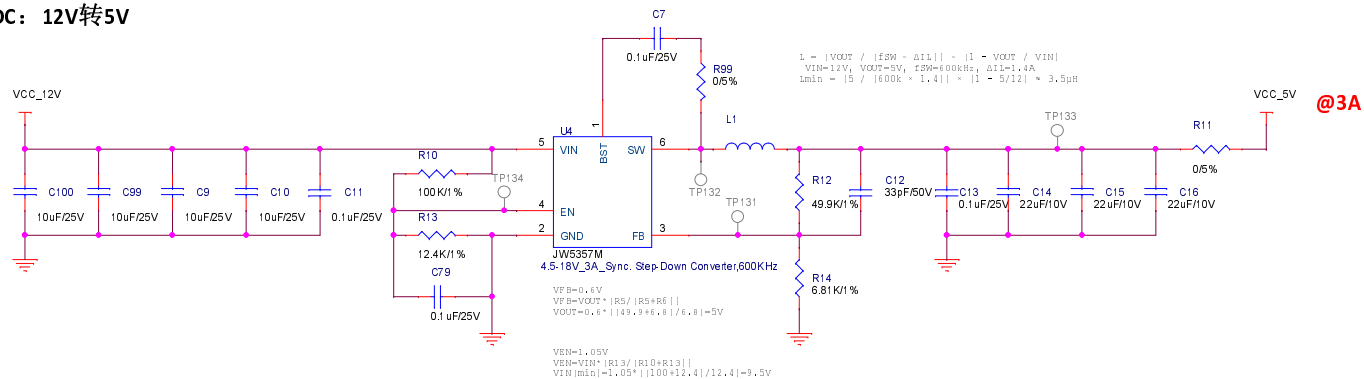
BOOT



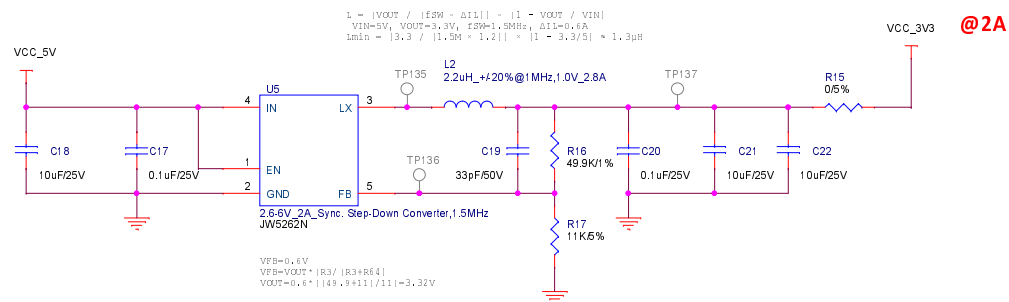
电平转换



DCDC: 12V转5V



DCDC: 5V转3V3



V0.1

		Zhejiang Mairuike Technology Co., Ltd.			
		Design Name			
		T28_Upper Control Board			
		Size A3	Page Name		POWER
				Rev <V01>	
Date:		Thursday, June 18, 2026		Sheet	4 of 10

分页标识



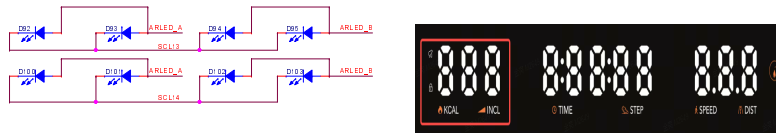
WiFi

静音

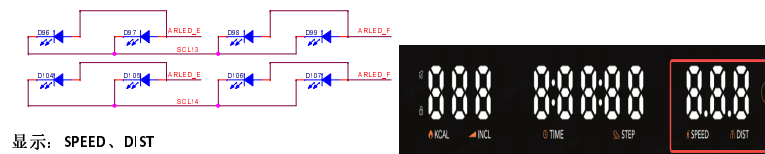
BT

童锁

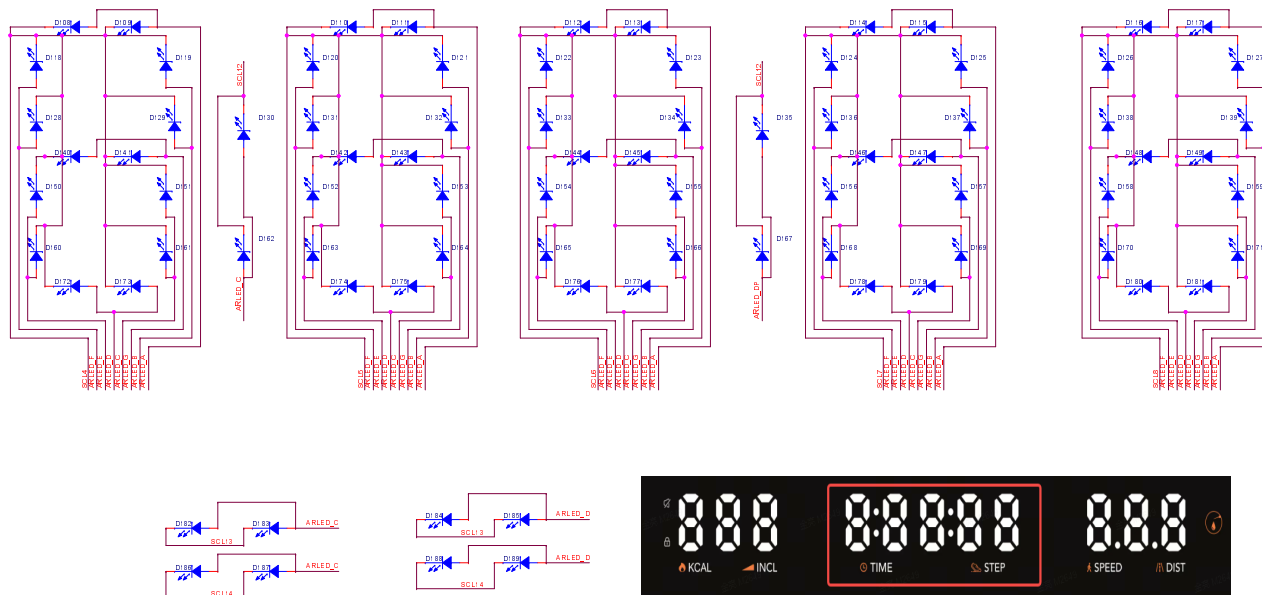
显示: KCAL、INCL



显示: SPEED、DIST



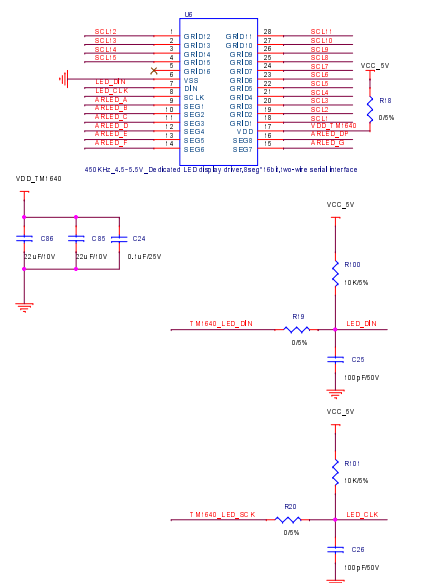
加油



显示: TIME、STEP



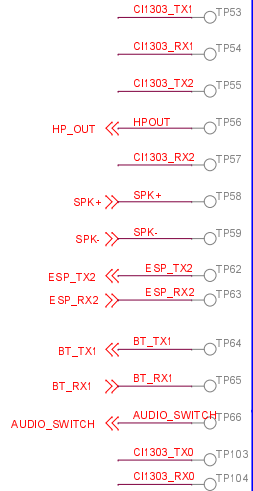
LED断码驱动



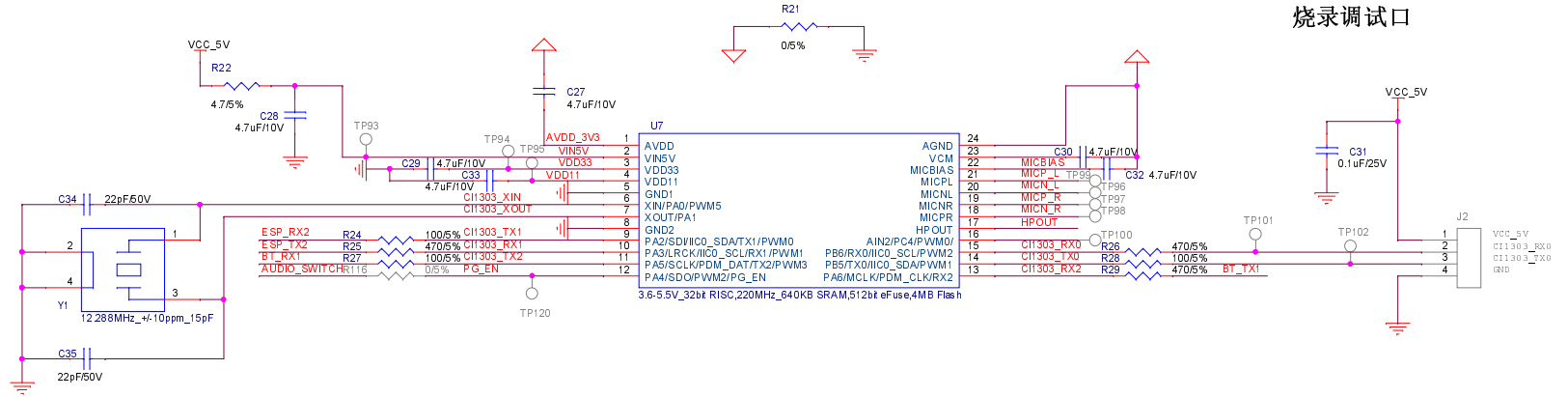
VD.1

Zhongxing Technology Co., Ltd.			
Design Name	T28_Upper Control Board		
Drawn	Page Name	LED	Rev
Drawn	Thursday, June 18, 2020	Sheet	5 of 15

分页标识

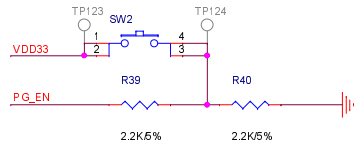


CI1303



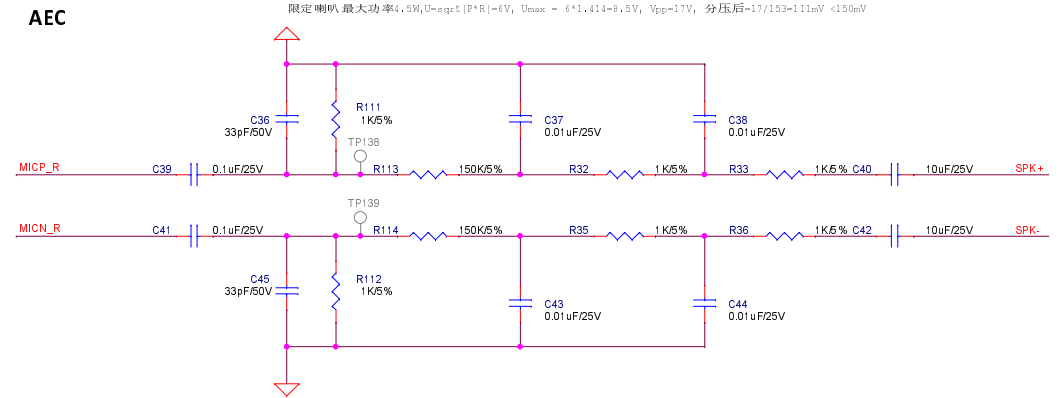
烧录调试口

BOOT按键

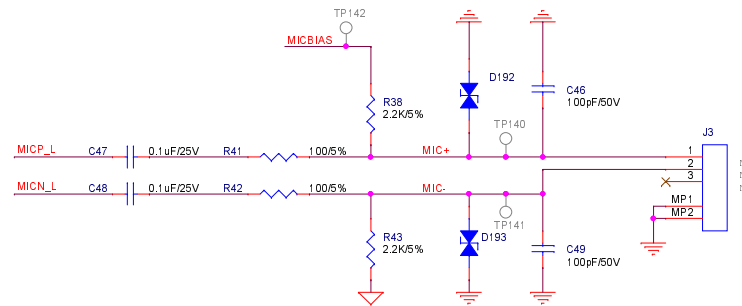


Boot Mode	PG_EN
升级模式	1
普通模式	0

AEC



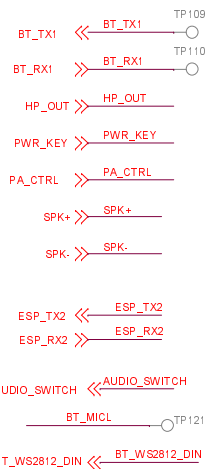
麦克风



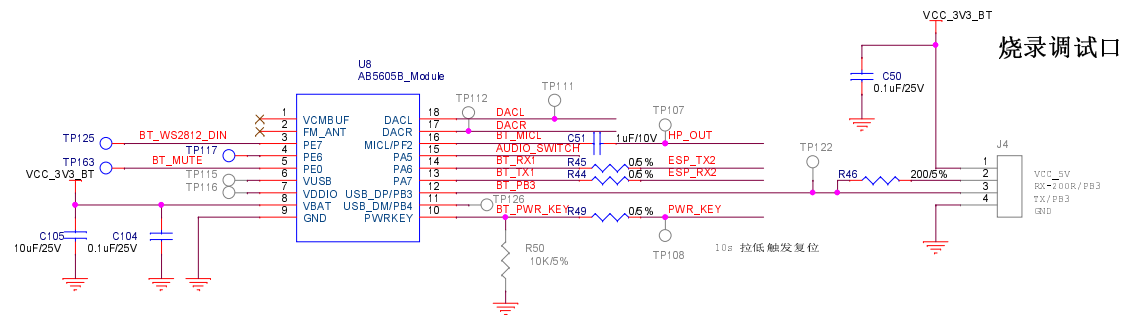
V0.1

Zhejiang Mairuike Technology Co., Ltd.			
Design Name	T28_Upper Control Board		
Size A3	Page Name	AUDIO	
Date	Thursday, June 18, 2026	Sheet	6 of 10

分页标识

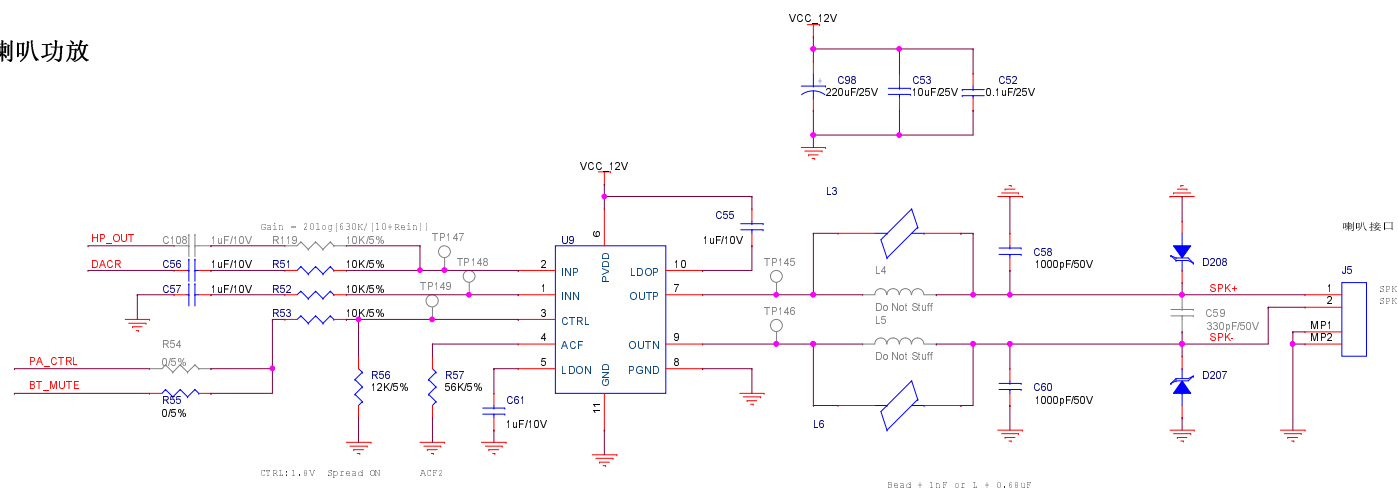


AB5605

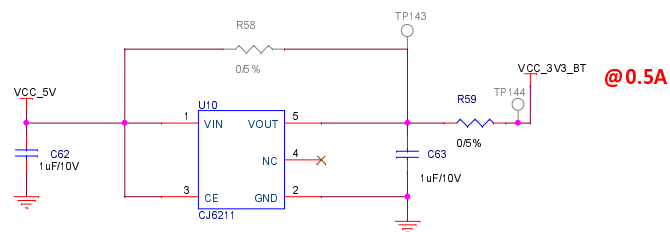


烧录调试口

喇叭功放



LDO: 5V转3.3V



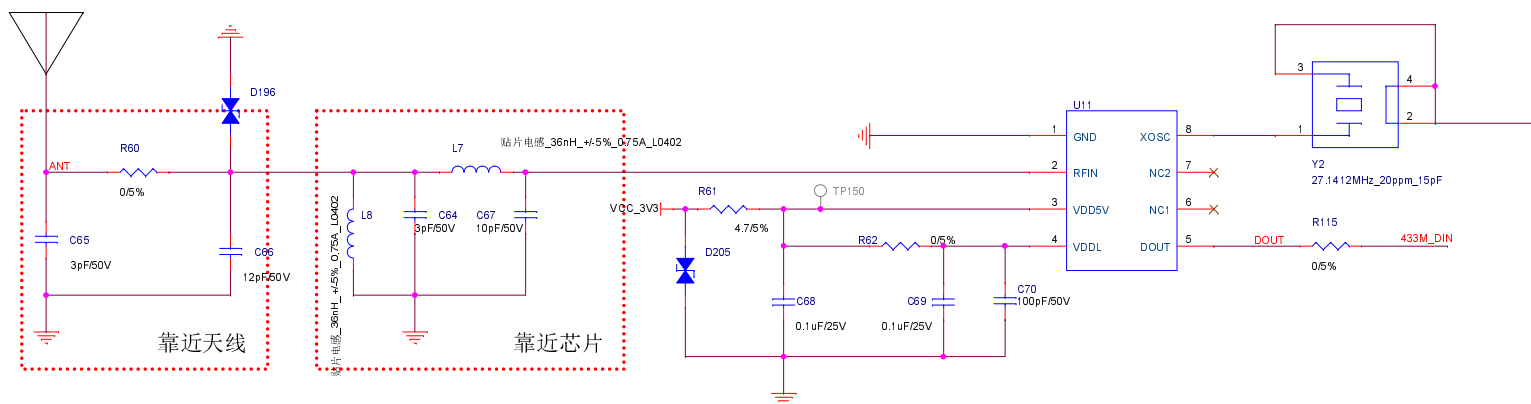
V0.1

			
Zhejiang Mairuike Technology Co., Ltd.			
Design Name			
T28_Upper Control Board			
Size			
A3			
Page Name			
BT MODULE			
Rev			
<V01>			
Date			
Monday, June 22, 2026			
Sheet			
7 of 10			

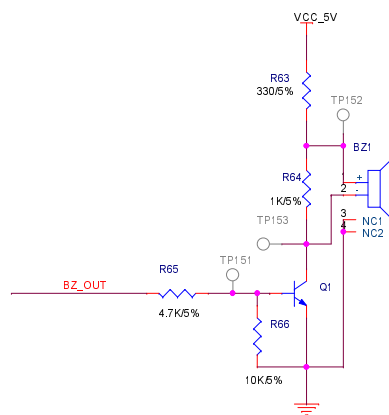
分页标识

433M接收

BZ_OUT >> BZ_OUT
433M_DIN << 433M_DIN



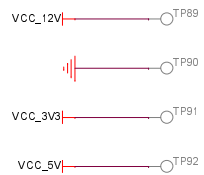
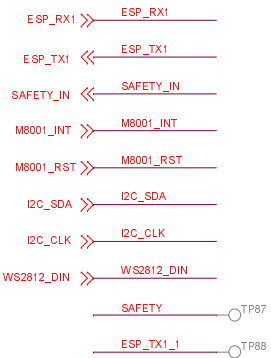
蜂鸣器



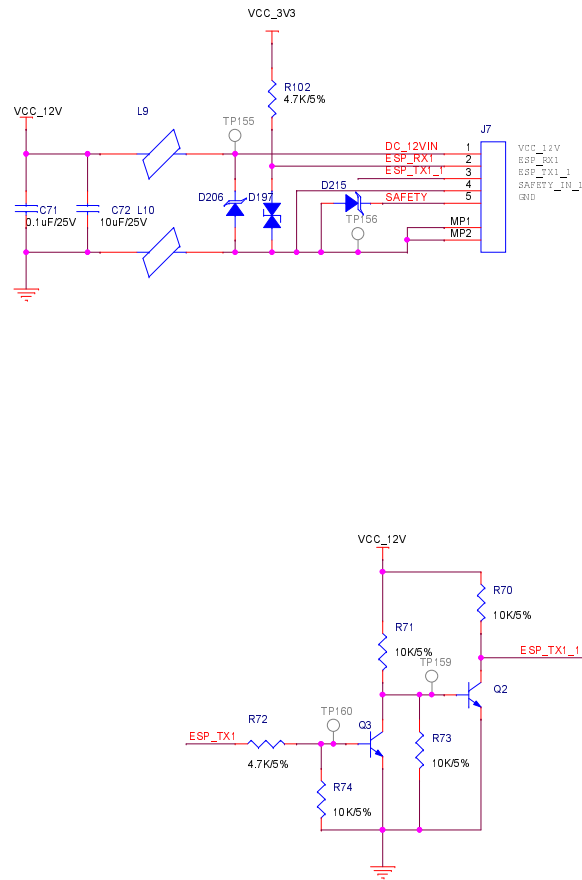
V0.1

Zhejiang Mairuike Technology Co., Ltd.			
Design Name T28_Upper Control Board			
Size A3	Page Name RF	Rev <V01>	
Date: Thursday, June 18, 2026	Sheet 8 of 10		

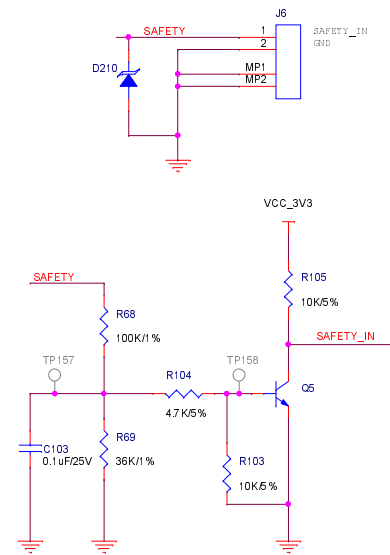
分页标识



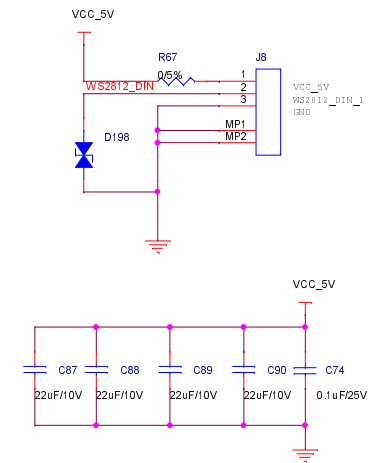
下控板接口



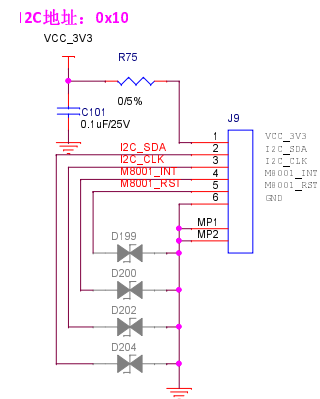
安全锁接口



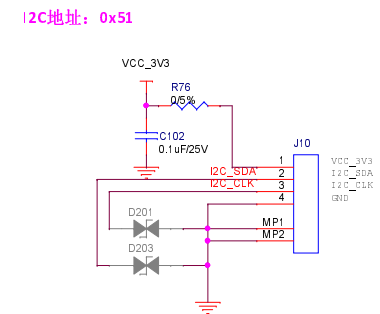
灯带接口



测距传感器接口



RTC小板接口



V0.1

			
Zhejiang Mairuike Technology Co., Ltd.			
Design Name T28_Upper Control Board			
Size A3	Page Name INTERFACE		Rev <v01>
Date	Thursday, June 18, 2026	Sheet 9	of 10

D	LCD_CS_n	>>>	LCD_CS_n
	LCD_RS	>>>	LCD_RS
	LCD_WR_n	>>>	LCD_WR_n
	LCD_RD_n	>>>	LCD_RD_n
	LCD_RST_n	>>>	LCD_RST_n
	LCD_D0	>>>	LCD_D0
	LCD_D1	>>>	LCD_D1
	LCD_D2	>>>	LCD_D2
	LCD_D3	>>>	LCD_D3
	LCD_D4	>>>	LCD_D4
C	LCD_D5	>>>	LCD_D5
	LCD_D6	>>>	LCD_D6
	LCD_D7	>>>	LCD_D7
	BL_PWM	>>>	BL_PWM
	TP_INT	>>>	TP_INT
	TP_RST	>>>	TP_RST
	I2C_SDA	>>>	I2C_SDA
	I2C_CLK	>>>	I2C_CLK

I2C地址: 0x5D or 0x14

